

FINAL INTERNSHIP TECHNICAL REPORT

Renewable Energy & Solar PV Technology

Technical Study of Geothermal Energy, MHD Systems, Direct Thermal Energy Conversion and Ocean Thermal Energy Conversion (OTEC)

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Internship Track	Renewable Energy & Solar PV Technology
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Evaluation Period	30/08/2026 – 30/09/2026

Prepared for Final Internship Evaluation
Academic Year 2026

Submission note: The GitHub repository URL must be inserted in the designated section before final submission.

INTERNSHIP COMPLETION CERTIFICATE — SUBMISSION TEMPLATE

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Certificate / verification reference: _____

Date of issue: _____

Authorized signatory: _____

Organization seal: _____

Student signature	Corporate mentor / authorized signatory
_____	_____

DECLARATION

I hereby declare that this internship technical report has been prepared as part of my corporate internship and academic learning in the field of renewable energy and thermal energy conversion. The report consolidates the technical concepts documented during the internship, including geothermal energy systems, magnetohydrodynamic (MHD) power generation, thermodynamic performance, exergy, direct thermal/electrochemical energy conversion, and Week 4 learning on Ocean Thermal Energy Conversion (OTEC).

The technical discussion is presented for educational and internship-evaluation purposes. Equations and engineering relationships are stated in standard form, while practical design values and operating ranges should be verified against detailed plant data, applicable standards, and manufacturer specifications before engineering implementation.

Student	Corporate Mentor / Supervisor
Ajay Prakash Goswami	Vikash
Signature: _____	Signature: _____
Date: _____	Date: _____

ACKNOWLEDGEMENT

I express my sincere gratitude to Rajkiya Engineering College Banda for providing the academic environment and opportunity to undertake this internship. I am thankful to my corporate mentor, Vikash, Technical Instructor at Nexonix Reflexo, for guidance and technical direction during the internship period.

I also acknowledge the contribution of faculty members, technical resources, and learning materials that supported my understanding of geothermal energy conversion, MHD systems, direct thermal energy conversion, fuel cells, and Ocean Thermal Energy Conversion. The internship helped connect thermodynamics, fluid mechanics, heat transfer, electromagnetic conversion, materials, and renewable-energy concepts with practical energy-system architecture.

Finally, I am grateful to everyone who supported the preparation, review, organization, and verification of this final internship report.

Student: Ajay Prakash Goswami

Programme: B.Tech Mechanical Engineering

Institution: Rajkiya Engineering College Banda

ABSTRACT

This report presents a structured technical study developed during the Renewable Energy & Solar PV Technology internship. The major technical areas documented are geothermal energy, magnetohydrodynamic power generation, direct thermal/electrochemical energy conversion, and Week 4 Ocean Thermal Energy Conversion (OTEC).

The geothermal study covers resource types, enhanced geothermal systems, thermodynamic limits, Carnot efficiency, exergy, non-condensable gases, dry-steam, flash-steam and binary/Organic Rankine Cycle pathways, direct-use applications, and cascaded utilization. The MHD study explains the Lorentz-force principle, conductivity, velocity, magnetic field, open- and closed-cycle arrangements, plasma seeding, Hall effects, electrode erosion, superconducting magnets, and seed recovery. Direct conversion is discussed through thermoelectric, thermionic, MHD and fuel-cell technologies.

The Week 4 OTEC evidence documents closed-cycle operation, engineering developments, techno-economic considerations, and future applications. Overall, the internship strengthened an integrated understanding of renewable thermal resources, thermodynamic limits, direct conversion principles, engineering constraints, and system-level energy utilization.

Keywords: geothermal energy, MHD, exergy, Carnot efficiency, flash steam, binary cycle, ORC, EGS, thermoelectric, thermionic, fuel cell, OTEC, LCOE.

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1. INTERNSHIP PROFILE AND PROJECT SCOPE

The uploaded internship report identifies the corporate internship track as Renewable Energy & Solar PV Technology and records the evaluation period as 30/08/2026–30/09/2026. The technical report is organized as a study of renewable thermal resources and energy-conversion technologies rather than as a single fabricated hardware prototype.

Item	Recorded detail
Student	Ajay Prakash Goswami
Programme	B.Tech in Mechanical Engineering
College	Rajkiya Engineering College Banda
Internship track	Renewable Energy & Solar PV Technology
Intern UID	NXR/INT/2026/063
Corporate mentor	Vikash — Technical Instructor, Nexonix Reflexo
Evaluation period	30/08/2026 – 30/09/2026

The project scope represented in this final report is: analysis and technical documentation of renewable thermal energy resources and direct/advanced energy-conversion systems, supported by internship learning evidence.

2. OBJECTIVES AND TECHNICAL METHODOLOGY

The documented internship objectives are to understand geothermal resources; apply Carnot efficiency, entropy and exergy; compare geothermal electricity-generation pathways; study MHD operating principles and cycle arrangements; understand direct thermal/electrochemical conversion; and identify major engineering challenges such as corrosion, scaling, non-condensable gases, electrode erosion, seed recovery and material durability.

The report was structured by consolidating the supplied internship learning material into technical chapters, comparing conversion pathways, recording key equations, and preserving presentation evidence as figures. The Week 4 OTEC evidence was integrated into the main report rather than left as an after-report supplement.

Energy-conversion chain used as the organizing concept: Primary heat source → working fluid → energy conversion → electrical/thermal output → end use.

3. GEOTHERMAL ENERGY: RESOURCE FOUNDATIONS

Geothermal energy is usable thermal energy stored within the Earth. The documented resource forms include hot water, steam, hot fractured rock, geopressured fluids, and very high-temperature magmatic systems. Engineering value depends on temperature, pressure, flow rate, permeability, fluid chemistry, depth, reinjection conditions and reservoir sustainability.

Resource type	Main characteristic	Engineering use
Hydrothermal reservoir	Natural heat source, permeable reservoir, water/steam fluid	Direct use, steam/flash/binary power
Geopressured brine	Deep, highly pressurized brine; may contain dissolved gases	Heat and fluid-energy recovery
Hot dry/fractured rock	High-temperature formation with limited natural permeability	Engineered geothermal systems
Magma resource	Extremely high-temperature thermal resource	Emerging/experimental concept

Enhanced Geothermal Systems (EGS): EGS seeks to create or enhance a permeable heat-exchange volume in hot rock where natural hydrothermal permeability is insufficient. Directional drilling, hydraulic stimulation, fracture characterization, circulation control and reinjection are key considerations.

4. THERMODYNAMIC FOUNDATIONS AND EXERGY

Geothermal resources often operate at lower temperatures than conventional fossil-fuel thermal plants. Therefore, the maximum theoretical thermal efficiency is limited by the source and sink temperatures.

Concept	Relationship / Interpretation
Carnot efficiency	$\eta_{\text{Carnot}} = 1 - T_0/T_g$; temperatures must be in kelvin.
Flow exergy	$e_x = (h - h_0) - T_0(s - s_0)$
NCG effect	Non-condensable gases can raise condenser pressure and increase gas-removal work, reducing net output.

Exergy measures the maximum theoretical useful work obtainable as a system moves toward equilibrium with the environment. In geothermal systems, this is important because a large quantity of low-temperature heat can contain comparatively limited work potential.

Actual plant efficiency is lower than the reversible limit because of irreversibilities, pressure losses, heat-transfer limitations, turbine/generator losses, parasitic loads and resource constraints.

5. GEOTHERMAL ELECTRICAL CONVERSION PATHWAYS

System	Primary conversion	Typical condition	Key feature
Dry steam	Steam → turbine → generator	Steam-dominated resource	Simple thermodynamic path
Flash steam	Hot brine → pressure drop → steam → turbine	High-temperature liquid-dominated resource	Steam produced by flashing
Binary / ORC	Geothermal fluid → heat exchanger → secondary fluid	Moderate-temperature resource	Closed secondary working loop

In a dry-steam system, steam is supplied to a turbine after appropriate separation and conditioning. In a flash-steam system, pressure reduction partially vaporizes hot pressurized geothermal water. Binary/ORC systems transfer geothermal heat through a heat exchanger to a secondary fluid with a lower boiling temperature, keeping the geothermal fluid separated from the power-cycle working fluid.

Technology selection depends on resource temperature and phase, pressure, chemistry, non-condensable gases, well productivity, cooling conditions, environmental requirements and project economics.

6. DIRECT USE AND CASCADED GEOTHERMAL SYSTEMS

Geothermal heat does not always need to be converted into electricity. Direct use can provide higher overall utilization when the required temperature matches the resource. Documented applications include space heating, district heating, greenhouse heating, aquaculture, bathing, drying, food processing and selected industrial processes.

Temperature level	Possible use
High	Electricity generation / high-temperature process heat
Medium	Industrial drying, process heating, greenhouse applications
Low	Space heating, hot water, aquaculture
Very low	Heat pumps / low-grade heating applications

Cascaded utilization uses the same geothermal fluid or extracted heat at progressively lower temperature levels. This can improve the fraction of the resource that is productively utilized.

Key challenges: drilling cost, resource uncertainty, scaling, corrosion, induced-seismicity concerns in some engineered systems, and long-term reservoir management.

7. MHD SYSTEM FUNDAMENTALS

Magnetohydrodynamics combines fluid mechanics and electromagnetism. When an electrically conducting fluid moves through a magnetic field, charged particles experience the Lorentz force. Charge separation produces an electromotive force that can drive an external current.

Parameter	Role in MHD conversion
Conductivity, σ	Controls the ability of the fluid/plasma to carry electrical current.
Velocity, v	Higher velocity increases electromagnetic interaction and power-density potential.
Magnetic flux density, B	Strong magnetic fields increase conversion potential but add structural/cooling demands.
Loading factor, K	Represents electrical loading in the simplified power-density relation.

Lorentz force: $F = q(E + v \times B)$

Simplified power-density scaling: $P_d \propto \sigma v^2 B^2 K(1-K)$

8. MHD CYCLES, WORKING FLUIDS AND DESIGN CHALLENGES

Parameter	Open cycle	Closed cycle
Working fluid	Combustion gas / plasma	Recirculating inert gas or controlled fluid
Seed management	Injection and recovery may be required	Controlled addition/recovery within loop
Heat source	Often direct combustion	Indirect heating possible
Main concern	Corrosion, electrode/channel interaction	Loop complexity and heat-transfer requirements

Plasma ionization is required for useful electrical conductivity. Alkali-metal seeding can increase electron availability and conductivity at more manageable engineering temperatures. The documented design challenges include Hall effects, electrode erosion, superconducting-magnet requirements, seed recovery and high-temperature material degradation.

Engineering response directions: improved materials, current-density control, cooling, channel/electrode design, magnetic-field control, efficient seed recovery, advanced magnet support systems, and refractory thermal-management approaches.

9. DIRECT THERMAL AND ELECTROCHEMICAL ENERGY CONVERSION

Technology	Conversion principle	Important parameter
MHD	Moving conducting fluid in a magnetic field	Conductivity, velocity, magnetic field
Thermoelectric	Seebeck effect across a temperature gradient	Seebeck coefficient, conductivity, thermal conductivity
Thermionic	Electron emission from a heated surface	Emitter temperature, work function, space charge
Fuel cell	Electrochemical conversion of chemical free energy	Cell voltage, current density, losses

Thermoelectric generators develop a voltage when a temperature difference is maintained across a suitable material and contain no rotating machinery. Their performance is often represented using the dimensionless figure of merit: $ZT = S^2\sigma T/k$. Thermionic conversion uses electrons emitted from a hot emitter and collected by a cooler electrode; high-temperature materials, space-charge effects, vacuum/cesium environments and heat rejection are important.

10. FUEL CELL FUNDAMENTALS AND VOLTAGE LOSSES

A fuel cell converts chemical free energy directly into electrical energy through electrochemical reactions. Its ideal efficiency is not constrained by the Carnot limit in the same way as a conventional heat engine because the primary conversion mechanism is electrochemical rather than thermal combustion followed by a heat engine.

Fuel-cell type	Operating feature / application emphasis
PEMFC	Polymer electrolyte; relatively low temperature; transport and distributed power
AFC	Alkaline electrolyte; specialized applications; sensitive to CO ₂
PAFC	Phosphoric acid; stationary CHP
MCFC	Molten carbonate; high temperature; stationary generation/CHP
SOFC	Solid oxide; high temperature; stationary and hybrid systems

Free-energy relation: $\Delta G = -nFE$

Actual voltage: $V_{\text{actual}} = E_{\text{rev}} - \eta_{\text{act}} - \eta_{\text{ohmic}} - \eta_{\text{conc}}$

The three practical losses are activation polarization, ohmic resistance and concentration/mass-transfer polarization.

11. WEEK 4 — OCEAN THERMAL ENERGY CONVERSION (OTEC)

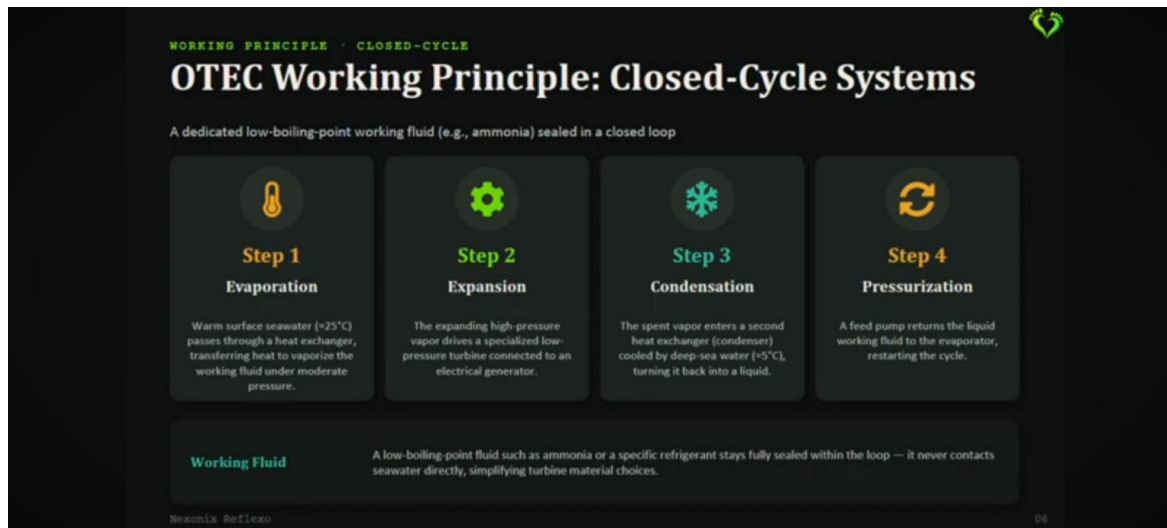
Week 4 focused on Ocean Thermal Energy Conversion (OTEC), specifically closed-cycle operation, engineering developments, economic performance and future applications. OTEC uses the temperature difference between warm surface seawater and cold deep-ocean water to drive a thermodynamic cycle and produce useful power.

- Explain evaporation, expansion, condensation and pressurization in a closed-cycle OTEC system.
- Understand why specialized heat exchangers, deep-water pipes and marine materials are important.
- Relate capital cost and levelized cost of energy (LCOE) to OTEC economics and recognize the potential value of multi-use platforms.
- Identify emerging applications including green-fuel production, seawater mineral recovery and floating sustainable infrastructure.

Topic	Week 4 focus
Working principle	Closed-cycle OTEC using a low-boiling-point working fluid
Engineering	Advanced materials, compact heat exchangers and anti-fouling methods
Economics	CAPEX, LCOE and multi-use revenue streams
Future scope	Green hydrogen/e-fuels, critical minerals and floating cities

12. OTEC TECHNICAL EVIDENCE AND ENGINEERING INTERPRETATION

Figure 1. OTEC working principle: closed-cycle system



Warm surface seawater supplies heat to vaporize the working fluid; the vapor expands through a turbine-generator; cold deep seawater condenses the vapor; and a pump returns the liquid to the evaporator.

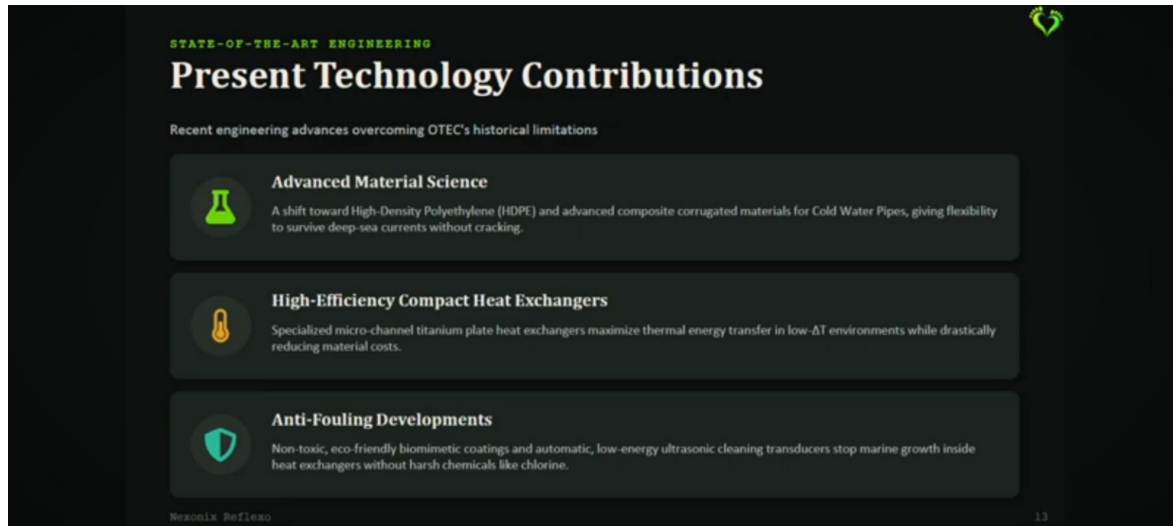
Figure 2. Economic performance and LCOE



The evidence records high initial capital requirements, comparatively high LCOE for pilot systems, and the potential for improved economics through desalination, seawater air conditioning (SWAC) and aquaculture co-benefits.

13. OTEC — ENGINEERING DEVELOPMENT AND FUTURE SCOPE

Figure 3. Present technology contributions



The engineering-development evidence highlights materials for cold-water pipes, compact titanium heat exchangers and anti-fouling approaches intended to improve durability, heat transfer and maintainability.

Figure 4. Applications and future scope



The evidence identifies potential OTEC-linked applications such as offshore green-fuel production, recovery of trace/critical minerals from seawater and integration with sustainable floating infrastructure.

14. LEARNING OUTCOMES

The report documents the following consolidated learning outcomes from the internship:

- Understanding the origin, classification and engineering characteristics of geothermal resources.
- Applying thermodynamic concepts such as Carnot efficiency and exergy to renewable thermal conversion.
- Comparing dry-steam, flash-steam and binary/ORC geothermal electricity-generation pathways.
- Understanding MHD conversion through conductivity, fluid velocity, magnetic fields, loading and Lorentz-force interaction.
- Recognizing practical MHD limitations including Hall effects, electrode erosion, seed recovery, high-field magnets and material degradation.
- Understanding direct thermal/electrochemical conversion through thermoelectric, thermionic, MHD and fuel-cell technologies.
- Understanding closed-cycle OTEC operation, marine engineering requirements, techno-economic considerations and future multi-use applications.

15. DOCUMENTED INTERNSHIP ACHIEVEMENTS

The following achievements are supported by the contents and evidence contained in the submitted report; they are stated as documented internship outputs rather than external awards or certifications.

Achievement / output	Evidence in final report
Structured technical report	Integrated chapters covering geothermal, MHD, direct conversion, fuel cells and OTEC.
Technical comparison	Comparison tables for geothermal pathways, MHD cycle arrangements and direct-conversion technologies.
Engineering formula sheet	Carnot efficiency, flow exergy, Lorentz force, MHD power-density scaling, ZT and fuel-cell relations.
Presentation/learning evidence	Four OTEC technical-evidence figures integrated into the final report.
Systems perspective	Integrated discussion of renewable heat, direct use, electrical conversion, direct-conversion devices and future energy systems.

16. TECHNICAL SCREENSHOTS / PRESENTATION EVIDENCE

The following figures are extracted from the supplied internship presentation/report evidence. They are retained as evidence of the technical learning material incorporated into the report.

Geothermal resources and direct-conversion evidence



Figure 5. Geothermal resources: origins and typologies



Figure 6. Thermionic systems: challenges and engineering barriers

Additional comparative and advanced-material evidence

COMPARATIVE ANALYSIS		
Performance Matrix: TE vs. Thermionics		
Head-to-head technical and operational comparison across key engineering parameters		
	THERMOELECTRICS	THERMIONICS
Conversion Efficiency	5-8% (ZT=1 limit)	10-20% (topping cycle)
Operating Temperature	300 K - 1,200 K	1,500 K - 2,500 K (emitter)
Power Density	0.1-1 W/cm²	10-100 W/cm²
Working Fluid	None — solid state	Vacuum or Cesium vapor
Moving Parts	Zero — fully static	Zero — fully static
Primary Weakness	Low efficiency ceiling	Extreme temp + corrosion
Best Application	RTGs, waste heat, IoT	Nuclear/solar topping cycles

Figure 7. Performance matrix: thermoelectrics versus thermionics

RESEARCH FRONTIERS · NANOMATERIALS & NEXT-GEN

Advanced Materials: Nanostructures & MEMS

Cutting-edge engineering bridging material science, quantum effects, and aerospace power systems

Sub-µm

Vacuum gap precision

MEMS Fabrication

Microscopic silicon cantilever architectures maintain stable sub-micron vacuum gaps without corrosive cesium gas assistance — enabling compact thermionic microsystems.

< 1.0 eV

Carbon nanotube work function

Graphene & CNT Collectors

Graphene-coated collectors and carbon nanotube arrays reduce collector work functions below 1.0 eV while simultaneously dropping back-radiation thermal footprints.

10-20%

Topping cycle efficiency

Nuclear & Solar Topping

High power densities make thermionics an excellent topping cycle for space nuclear reactor cores and concentrated solar farms — cascading with Rankine/Stirling bottoming.

Figure 8. Advanced materials: nanostructures and MEMS

17. GITHUB SUBMISSION — REQUIRED BEFORE FINAL SUBMISSION

The current uploaded report does not contain a verified GitHub repository URL. Because the assignment explicitly requires GitHub, this is the main item that must be completed before final submission.

GitHub item	Action
Repository name	Recommended: nexonix-reflexo-internship-report
Repository visibility	Public, unless the instructor specifically requires private access
Final report PDF	Upload the final PDF produced with this report
README.md	Describe internship, project scope, learning outcomes, evidence and files
Evidence folder	Upload relevant technical screenshots/figures
Repository URL	_____ (paste actual GitHub URL)

Recommended repository structure:

```
nexonix-reflexo-internship-report/  
├── README.md  
├── Ajay_Prakash_Goswami_Final_Internship_Report.pdf  
├── evidence/  
│   ├── OTEC_Working_Principle.jpg  
│   ├── OTEC_Economics_LCOE.jpg  
│   ├── OTEC_Technology.jpg  
│   └── OTEC_Future_Scope.jpg  
└── references/
```

Important: Do not submit a placeholder GitHub URL. Create the repository, upload the files, open the repository in a browser, and paste the real URL into the blank field above.

18. ENGINEERING CHALLENGES AND FUTURE SCOPE

Area	Key challenges documented
Geothermal	Drilling cost, reservoir uncertainty, scaling, corrosion, fluid chemistry, reinjection and sustainability.
EGS	Fracture creation/control, induced-seismicity management, long-term circulation and thermal breakthrough.
MHD	Plasma conductivity, electrode durability, Hall effects, high-field magnets, seed recovery and high-temperature materials.
Direct conversion	Material performance, temperature limits, heat rejection, power density and degradation.
Fuel cells	Catalyst durability, contamination, thermal/water management, cost, hydrogen production and infrastructure.
OTEC	High capital cost, heat-exchanger/pipe requirements, marine materials, anti-fouling and economic viability.

The documented future direction includes higher-temperature geothermal resources, advanced drilling, engineered reservoirs, improved heat-exchanger materials, hybrid geothermal systems, advanced MHD concepts, high-performance thermoelectrics, integrated hydrogen/fuel-cell systems, digital reservoir monitoring, predictive maintenance, advanced materials and system-level optimization. OTEC evidence additionally points toward green-fuel production, seawater mineral recovery and floating sustainable infrastructure.

19. CONCLUSION

The internship study demonstrates how renewable thermal resources can be connected to a broad family of energy-conversion technologies. Geothermal energy provides a continuous source of subsurface heat, while its conversion efficiency is constrained by source temperature and real engineering irreversibilities. Carnot efficiency provides the theoretical upper bound, while exergy analysis helps evaluate energy quality and identify useful-work destruction.

Dry-steam, flash-steam and binary/ORC systems provide different geothermal electricity pathways, while direct use and cascaded systems can improve total resource utilization. MHD generation illustrates direct electromagnetic conversion of an electrically conducting fluid, with performance strongly influenced by conductivity, velocity, magnetic-field strength, loading and channel design.

Direct conversion technologies such as thermoelectrics, thermionics, MHD and fuel cells demonstrate alternatives to the complete conventional heat-engine chain. Week 4 OTEC learning extended the renewable-thermal perspective to ocean temperature gradients, closed-cycle operation, marine engineering, LCOE and multi-use future applications.

Overall learning outcome: The internship provided an integrated understanding of renewable thermal resources, thermodynamic limits, direct energy-conversion principles, engineering constraints and the importance of matching energy quality and temperature level to useful end uses.

20. REFERENCES

1. DiPippo, R., Geothermal Power Plants: Principles, Applications, Case Studies and Environmental Impact, Butterworth-Heinemann.
2. Çengel, Y. A. and Boles, M. A., Thermodynamics: An Engineering Approach, McGraw-Hill.
3. O'Hayre, R., Cha, S.-W., Colella, W. and Prinz, F. B., Fuel Cell Fundamentals, Wiley.
4. Rogers, G. F. C. and Mayhew, Y. R., Engineering Thermodynamics, applicable editions.
5. Standard academic/technical resources on magnetohydrodynamics, plasma conductivity, electromagnetic energy conversion and thermal power systems.
6. Internship learning material and technical presentation material supplied during the Renewable Energy & Solar PV Technology internship.
7. Week 4 OTEC technical presentation evidence supplied as part of the internship learning material.

Source integrity note: This final report reformats and integrates the supplied internship report content. Unsupported personal achievements, project measurements, laboratory results, or external certificates have not been invented.

APPENDIX A — KEY FORMULAE

Topic	Formula
Carnot efficiency	$\eta = 1 - T_0/T_g$
Flow exergy	$e_x = (h - h_0) - T_0(s - s_0)$
Lorentz force	$F = q(E + v \times B)$
MHD power-density scaling	$P_d \propto \sigma v^2 B^2 K(1-K)$
Thermoelectric figure of merit	$ZT = S^2 \sigma T / k$
Fuel-cell free-energy relation	$\Delta G = -nFE$
Actual fuel-cell voltage	$V_{\text{actual}} = E_{\text{rev}} - \eta_{\text{act}} - \eta_{\text{ohmic}} - \eta_{\text{conc}}$

APPENDIX B — ABBREVIATIONS

Abbreviation	Meaning
MHD	Magnetohydrodynamics / Magnetohydrodynamic
EGS	Enhanced Geothermal System
HDR	Hot Dry Rock
ORC	Organic Rankine Cycle
NCG	Non-Condensable Gas
PEMFC	Proton Exchange Membrane Fuel Cell
AFC	Alkaline Fuel Cell
PAFC	Phosphoric Acid Fuel Cell
MCFC	Molten Carbonate Fuel Cell
SOFC	Solid Oxide Fuel Cell
CHP	Combined Heat and Power
OTEC	Ocean Thermal Energy Conversion
LCOE	Levelized Cost of Energy
SWAC	Seawater Air Conditioning

FINAL SUBMISSION CHECKLIST

- Insert the real GitHub repository URL in Section 17.

- Upload the final PDF to the GitHub repository.
- Upload the relevant evidence screenshots/figures.
- Verify that the official internship completion certificate is attached when issued.
- Open the final PDF once after download and check page order, signatures and GitHub URL before submission.